

Varun Maram

Assistant Professor

Department of Computer Science

University of Warwick

Research Interests

My interests lie in *quantum-resistant cryptography*, with an emphasis on provable post-quantum security of real-world cryptosystems. I also have a broader interest in *multi-party computation* and *distributed systems*. Recently, I've been interested in constructing advanced quantum-secure applications – such as anonymous credentials and verifiable oblivious PRFs – from zkSNARKs.

Education

- 2019 - **Swiss Federal Institute of Technology, Zurich (ETH Zurich), Zurich**
- 2023 Dr.Sc. in Computer Science, member of the Applied Cryptography Group
Supervisor: Kenneth G. Paterson

- 2017 - **Swiss Federal Institute of Technology, Zurich (ETH Zurich), Zurich**
- 2019 M.Sc. in Computer Science “with distinction”, GPA – **5.75/6.0**

- 2013 - **Indian Institute of Technology, Roorkee (IIT Roorkee), Roorkee**
- 2017 B.Tech. in Computer Science and Engg., minors in Mathematics, CGPA – **9.578/10**
(Departmental Rank 2, out of ≈ 75 students)

Work Experience

- 04/2026 - **University of Warwick, Coventry**
- Present *Assistant Professor* at the Department of Computer Science.

- 03/2026 - **Input Output (IOG), London**
- Present *Consultant*, Quantum Security.

- 03/2024 - **SandboxAQ, London**
- 03/2026 *Research Scientist – Postdoc*. Member of the Cybersecurity Group.
Manager(s): Nina Bindel, Andreas Hülsing
 - Currently analysing post-quantum versions of real-world protocols, and constructing advanced quantum-secure applications from zkSNARKs.

- 06/2022 - **Visa Research, Palo Alto**
- 09/2022 *Research Intern*. Worked on constructing quantum secure public-key encryption schemes. Mentor: Navid Alamati

- 06/2021 - **Visa Research, Palo Alto (worked remotely from Zurich)**
- 09/2021 *Research Intern*. Worked on analysing quantum (in)security of widely-used block

cipher modes of operation. Mentors: Daniel Masny and Sikhar Patranabis

- 05/2016 - **Adobe BigData Experience Lab**, *Bangalore*
07/2016 *Research Intern*. Conducted research on augmented reality technology in the context of digital marketing. Mentor: Gaurush Hiranandani

Standardization Efforts

- 2022 D. J. Bernstein, T. Chou, C. Cid, J. Gilcher, T. Lange, Varun Maram, I. von Maurich, R. Misoczki, R. Niederhagen, E. Persichetti, C. Peters, N. Sendrier, J. Szefer, C. J. Tjhai, M. Tomlinson, and W. Wang.
“Classic McEliece”, Round 4 Submission in NIST’s *Post-Quantum Cryptography (PQC) Standardization Project*, currently being considered for ISO standardization.

Publications

- 2026 Y-H. Huang, A. Hülsing, Varun Maram, S. Ritsch, A. Saha. **“Feistel Tools: Reprogramming and Query-Recording for QRPs”** [In Submission]
- 2025 N. Alapati, Varun Maram. **“Quantum CCA-Secure PKE, Revisited”**, *Journal of Cryptology – JoC*, Volume 38.
- 2025 B. Benčina, B. Dowling, Varun Maram, and K. Xagawa. **“Post-quantum Cryptographic Analysis of SSH”**, *46th IEEE Symposium on Security and Privacy 2025*.
[This work was also presented at the Real World Crypto (RWC) 2025 symposium.]
- 2024 N. Alapati, Varun Maram. **“Quantum CCA-Secure PKE, Revisited”**, 27th IACR Intl. Conference on Practice and Theory of Public-Key Cryptography – *PKC 2024*.
[Best Paper Award], [Invited to Journal of Cryptology]
- 2023 M. Jauch, Varun Maram. **“Quantum Cryptanalysis of OTR and OPP: Attacks on Confidentiality, and Key-Recovery”**, *Selected Areas in Cryptography – SAC 2023*.
- 2023 N. Alapati, Varun Maram, D. Masny. **“Non-Observable Quantum Random Oracle Model”**, 14th Intl. Conference on Post-Quantum Cryptography – *PQCrypto 2023*.
- 2023 Varun Maram, K. Xagawa. **“Post-Quantum Anonymity of Kyber”**, 26th IACR Intl. Conference on Practice and Theory of Public-Key Cryptography – *PKC 2023*.
[Best Paper Award]
- 2022 Varun Maram, D. Masny, S. Patranabis, and S. Raghuraman. **“On the Quantum Security of OCB”**, *IACR Transactions on Symmetric Cryptology, ToSC 2022 (2)*.
- 2022 P. Grubbs, Varun Maram, and K. G. Paterson. **“Anonymous, Robust Post-Quantum Public Key Encryption”**, 41st Annual Intl. Conference on the Theory and Applications of Cryptographic Techniques – *EUROCRYPT 2022*.
[This work was also presented at NIST’s 3rd PQC Standardization Conference.]
- 2021 K. Cong, D. Cozzo, Varun Maram, and N. P. Smart. **“Gladius: LWR Based Efficient**

Hybrid Public Key Encryption with Distributed Decryption”, 27th Annual Intl. Conference on the Theory and Application of Cryptology and Information Security – ASIACRYPT 2021.

- 2020 Varun Maram. **“On the Security of NTS-KEM in the Quantum Random Oracle Model”**, 8th Intl. Workshop on Code-Based Cryptography – *CBCrypto 2020*.
- 2020 C. Liu-Zhang, Varun Maram, and U. Maurer. **“On Broadcast in Generalized Network and Adversarial Models”**, 24th Conference on Principles of Distributed Systems – *OPODIS 2020*.
- 2019 C. Liu-Zhang, Varun Maram, and U. Maurer. **“Brief Announcement: Towards Byzantine Broadcast in Generalized Communication and Adversarial Models”**, 33rd International Symposium on Distributed Computing – *DISC 2019*.
- 2017 G. Hiranandani, K. Ayush, A. R. Sinha, Sai Varun Reddy Maram, C. Varsha, and P. Maneriker. **“[Poster] Enhanced Personalized Targeting Using Augmented Reality”** 16th IEEE Intl. Symposium on Mixed and Augmented Reality – *ISMAR 2017*.

Selected Talks

- 2025 **“Post-quantum Cryptographic Analysis of SSH”**
- Real World Crypto Symposium – *RWC 2025*
 - Secure Shell Maintenance (SSHM) Working Group Session – *IETF 123*
- 2024 **“Don’t Roll Your Own FO: Challenges in Proving Post-Quantum CCA Security of Real-World KEMs”**
- Workshop on Key Exchange and Channel Protocols – *SKECH 2024*
- 2024 **“Quantum CCA-Secure PKE, Revisited”**
- Information Security Group (ISG) Research Seminar, Royal Holloway (RHUL)
 - 27th IACR Intl. Conference on Practice and Theory of PKC – *PKC 2024*
- 2023 **“Post-Quantum Anonymity of Kyber”**
- CWI Amsterdam Crypto Group Seminar
 - 26th IACR Intl. Conference on Practice and Theory of PKC – *PKC 2023*
- 2023 **“Non-Observable Quantum Random Oracle Model”**
- 14th Intl. Conference on Post-Quantum Cryptography – *PQCrypto 2023*.
- 2023 **“On the Quantum Security of OCB”**
- Fast Software Encryption Conference – *FSE 2023*
- 2022/23 **“Anonymous, Robust Post-Quantum Public Key Encryption”**
- IBM Research Zurich, Security Department Seminar
 - ENS Lyon/CWI Amsterdam/RHUL Joint Online Cryptography Seminar
 - QSIT/Quantum Centre, ETH Zurich, Lunch Seminar
 - NIST’s Third PQC Standardization Conference
 - 41st Annual Conference on Cryptographic Techniques – *EUROCRYPT 2022*.

- 2020 **“On the Security of NTS-KEM in the Quantum Random Oracle Model”**
 - 8th Intl. Workshop on Code-Based Cryptography – *CBCrypto 2020*.
- 2020 **“On Broadcast in Generalized Network and Adversarial Models”**
 - 24th Conference on Principles of Distributed Systems – *OPODIS 2020*.
- 2019 **“Brief Announcement: Towards Byzantine Broadcast in Generalized Communication and Adversarial Models”**
 - 33rd International Symposium on Distributed Computing – *DISC 2019*.

Selected Honors and Awards

- 2024 **Best Paper Award** given by the program committee of PKC 2024 conference for our paper *“Quantum CCA-Secure PKE, Revisited”*; given to 2 papers out of 178 submissions.
- 2023 **Best Paper Award** given by the program committee of PKC 2023 conference for our paper *“Post-Quantum Anonymity of Kyber”*; given to 2 papers out of 183 submissions.
- 2017 **Master Scholarship** awarded by the Department of Computer Science, ETH Zurich to admitted students based on their excellent academic achievements so far; was offered to ≈ 10 students that year.
- 2014 **Aditya Birla Group Scholarship** awarded to engineering, management and law students all over India, based on their excellence on the academic and leadership front; given to ≈ 40 graduates in the country.
- 2012 **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship** awarded by Dept. of Science and Technology, Govt. of India, to highly motivated students interested in pursuing a research career; given to ≈ 250 high school students in the country.
- 2011 **Bronze Medal**, 16th International Astronomy Olympiad, Kazakhstan. One of the 3 high school students selected to represent India in the competition.
- 2011 **Infosys Award** for International Olympiad Medallists, by HBCSE in association with Tata Institute of Fundamental Research (TIFR) Endowment Fund
- 2011 **Gold Medal**, 13th National Science Olympiad, Science Olympiad Foundation, Delhi. National Rank 1st out of $\approx 20,000$ participants.

Teaching

I was a (lead) teaching assistant for the following courses at the Department of Computer Science, ETH Zurich.

- 2022-2023 **Zero-Knowledge Proofs** (*Master’s level*)
- 2023 **Seminar on Current Topics in Cryptography** (*Master’s level*)
- 2022 **Information Security** (*Bachelor’s level*)
- 2020-2021 **Applied Cryptography** (*Master’s level*)

2020-2021 **Algorithmic Game Theory** (*Master's level*)

Advising

I supervised the following theses at the Department of Computer Science, ETH Zurich.

- 2023 Melanie Jauch. **"Quantumania: Three Quantum Attacks on AES-OTR's Confidentiality and a Quantum Key-Recovery Attack on OPP"** (*Bachelor's thesis*)
[This work was also published at the SAC 2023 conference.]
- 2022 Marc Himmelberger. **"Concrete IND-CCA Security of NIST PQC KEMs in the ROM"** (*Bachelor's thesis*)
- 2021 Oliver Tran. **"Exploring RSA Assumptions"** (*Bachelor's thesis*)

Academic Service

I was/am on the program committee (resp., editorial board) for the following conferences (resp., journals):

- PKC 2027
- CiC 2026
- PKC 2025

I was also a sub-reviewer for the following conferences:

- CRYPTO 2026, EUROCRYPT 2026, PKC 2026
- CRYPTO 2025, ASIACRYPT 2025, TCC 2025
- CRYPTO 2024, ASIACRYPT 2024
- EUROCRYPT 2023
- CRYPTO 2021, ASIACRYPT 2021
- CRYPTO 2020, CT-RSA 2020

Patents

- 2016 G. Hiranandani, Sai Varun Reddy Maram, K. Ayush, C. Varsha, and S. Jain. **"Method, Medium, and System for Product Recommendations Based on Augmented Reality Viewpoints"**, US 10475103 B2.
- 2016 G. Hiranandani, C. Varsha, Sai Varun Reddy Maram, K. Ayush, and A. R. Sinha. **"Identifying Augmented Reality Visuals Influencing User Behavior in Virtual-Commerce Environments"**, US 10950060 B2.
- 2016 G. Hiranandani, K. Ayush, C. Varsha, and Sai Varun Reddy Maram. **"Creating Targeted Content Based on Detected Characteristics of an Augmented Reality Scene"**, US 10922716 B2.

Technical Skills

Programming C++, C, Python, Java, bash, C#

OS Linux (Ubuntu), macOS, Windows

Software Git, PostgreSQL, Docker, SageMath, liboqs, OpenSSH, EasyCrypt, LaTeX